representation in *SequenceFile* format from text documents under a directory structure.

```
./bin/mahout seqdirectory -i /news_article -o /news_article_
seqdir_output -ow

\-i  Path to job input directory. -o The directory pathname
for  output.  -ow  If present, overwrite the output directory
before running job
```

Now convert the text documents in the *SequenceFile* format to vectors using the following command:

```
./bin/mahout seq2sparse -i /news_article_seqdir_output -o /
news_article_sparse_output
```

If you issue the command *hadoop fs -ls /news_article_sparse_output*, you can see many files like *df-count, dictionary.file-0, frequency.file-0, f-vectors, tfidf-vectors, tokenized-document,* and *wordcount.*

The *dictionary* file contains the mapping between a term and its integer ID, and the other folders are intermediate folders generated during the vectorization process. TF-IDF stands for term frequency-inverse document frequency. The number of times a term occurs in a document is called its term frequency (tf); it is a ratio of the number of times the word occurs in the document to the total number of words in the document. The inverse document frequency (idf) is the log of the ratio of total number of documents and number of documents containing the term. tf*idf is a multiplication of tf and idf.

- The next step is to run the Kmeans clustering algorithm:

```
./bin/mahout kmeans -i /news_article_sparse_output/tfidf-
vectors -c /news_kmeans_cluster -o /news_kmeans_cluster_
output -x 2 -ow -k 50 -dm
```

This command takes the TF-IDF vector as an input, and *-c* input centroids as vectors. *-o* is the output directory, *-x* is the number of clusters, *–k* specifies that the centroids are randomly generated and are written to the input clusters folder, and *-dm* is distance measure (*SquaredEuclideanDistance-Measure* is set by default).

Mahout has a cluster dumper utility that can be used to retrieve and evaluate clustering data. Next, we need to run cluster dump:

```
./bin/mahout clusterdump -i /news_kmeans_cluster_output/
clusters-1-final -o /home/hduser/news_kmeans.txt -p /news_
kmeans_cluster_output/clusteredPoints -n 20
```

Here, *-i* is input (path to job input directory), *-o* is output directory pathname for output (it should point to your local file system and not the HDFS), *-p* is the directory containing points sequence files that map input vectors to their cluster (the program will output the points associated with the cluster), and *-n* is the number of top terms to print.

The generated output will have *"n": --,"* which is the number of elements in the cluster identifier; *c=[z, ...]:* is the centroid of the cluster, with the z's being the weights of the different dimensions; *r=[z, ...]:* is the radius of the cluster and the identifier signifying the clusters. This can again be programmatically converted for further interpretation of different clusters. END

References

[1] *https://mahout.apache.org/users/clustering/clusteringyourdata.html*

[2] Mahout in Action by Sean Owen, Robin Anil, Ted Dunning, Ellen Friedman ©2012 by Manning Publications Co. All rights reserved.

[3] *http://www.codeproject.com/Articles/439890/Text-Documents-Clustering-using-K-Means-Algorithm*

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